

Introduction to Optimal Control

Objectives:

We will be given a dynamic system, contain constraints, and a performance index to be minimized (or maximized).

We have to determine the control that achieves the objectives.

Dynamic system : equation of motion

$$\dot{\vec{x}} = f(\vec{x}, \vec{u}, t)$$

Constraints : state constraints, control constraints

Objective function or performance index cost

$$J(\vec{x}) = \phi(\vec{x}(t_f), t_f) + \int_0^{t_f} L(\vec{x}, \vec{u}, t) dt$$

ex.) $\dot{x} = f(x, u)$

$$J = \frac{1}{2} \int_0^{t_f} (x^2 + u^2) dt$$

$$\text{or } J = \frac{1}{2} \int_0^{t_f} (x^2 + 10u^2) dt$$

→ In this case, control is more important.

Minimum effort : $J = \int_0^{t_f} u^2 dt$

Minimum energy : $J = \int_0^{t_f} (\text{energy of the system}) dt$

Minimum fuel : $J = \int_0^{t_f} |u| dt$

Minimum time : $J = \int_0^{t_f} 1 dt = t_f$

$$J = \int_0^{t_f} 100 dt = 100t_f, \text{ same meaning}$$

Mayer problem : $J = \phi(x(t_f), t_f)$

Simple Optimal Control Problem (Example)

Minimize $J = \frac{1}{2} \int_0^{\infty} (x^2 + u^2) dt$

$$\dot{x} = u, \quad x(0) = 1 \text{ given} \quad \rightarrow \dot{x} = -kx$$

$$u = -kx$$

Find k !

This case is a linear quadratic problem.

Solution

$$\dot{x} = -kx$$

$$x(t) = x(0)e^{-kt}$$

$$\begin{aligned} J(k) &= \frac{1}{2} \int_0^{\infty} (x^2(0)e^{-2kt} + k^2 x^2(0)e^{-2kt}) dt \\ &= \frac{1}{2} x^2(0) \int_0^{\infty} e^{-2kt} (1 + k^2) dt \\ &= \frac{1}{2} x^2(0) (1 + k^2) \frac{1}{-2k} [e^{-2kt}]_0^{\infty} \\ &= \frac{1}{2} x^2(0) (1 + k^2) \frac{1}{2k} \\ &= \frac{1}{4} \frac{(1 + k^2)}{k} \end{aligned}$$

$$\frac{\partial J}{\partial k} = 0 \quad (\text{necessary condition})$$

$$\rightarrow \frac{1}{4} \frac{2k^2 - (1 + k^2)}{k^2} = 0$$

$$\Rightarrow k^2 = 1$$

$$\Rightarrow k = \pm 1$$

$$\frac{\partial^2 J}{\partial k^2} > 0 \quad (\text{sufficient condition for a minimum})$$

$$\frac{\partial^2 J}{\partial k^2} = \frac{\partial}{\partial k} \left(\frac{-1}{4k^2} + \frac{1}{4} \right) = \frac{1}{2k^3} > 0$$

$$\Rightarrow k^* = 1$$

$$\Rightarrow J^* = \frac{1}{2}$$

k is not function of time $\rightarrow$ parameter optimization

When u is not used and $\dot{x}$ is not restricted,

$$J = \frac{1}{2} \int_0^\infty (x^2 + \dot{x}^2) dt$$

$$x(0) = 1$$

This become a standard calculus of variations that does not treat control variables.

Optimal Control Problem

1) Establish an optimal control problem

performance index(measure)
dynamics system
constraints

2) Solve the problem

to determine a control function that minimizes or maximizes
the criterion (performance index)

Two strategies:

① The minimum (or maximum) principle (Pontryagin)

Nonlinear two-point boundary-value problem (TPBVP)

② Dynamic programming (Bellman)

Fundamental equation (Hamiltonian-Jacobi-Bellman equation).

In the method dynamic programming, an optimal policy is found by employing the intuitively appealing concept called the principle of optimality².

Many complex aerospace problems that are not amenable to classical techniques have been solved by using optimal theory. The optimal control law, if it can be obtained, usually requires a digital computer for implementation (except for the linear regulator problem), and all of the states must be available for feedback to the controller.

² Principle of optimality:

Whatever the initial state and initial decision are, the remaining decision must constitute an optimal policy with regard to the state resulting from the first decision.

Ex.) If a-b-e is the optimal path from a to e, the b-e is the optimal path from b to e.

The Calculus of Variations

<Functionals>

- A **function** f is a rule of correspondence that assigns each **element** $\vec{q}$ in a certain set $\mathbf{D}$ to a unique element in a set $\mathbf{R}$

$\mathbf{D}$: domain of f

$\mathbf{R}$: range of f

ex.) $f(\vec{q}) = \sqrt{q_1^2 + q_2^2 + \cdots + q_n^2}$

$q_1, \cdots, q_n$; coordinate of a point in n-dimensional Euclidean space.

$f(\vec{q})$ is the real number assigned by f is the distance of the point $\vec{q}$ from the origin.

- A **functional** J is rule of correspondence that assigns each **function** $\vec{x}$ in a certain set Ω to a unique real number $\mathbf{S}$

Ω : domain of J

$\mathbf{S}$: range of J

→ A functional is a “**function** of a **function**”

ex.) $J(x) = \int_{t_0}^{t_f} x(t) dt$
functional function

J is a unique real number.

J is the area under the $x(t)$ curve.

<Linearity of functional>

J is a linear functional if and only if it satisfies the principle of homogeneity

$$J(\alpha \vec{x}) = \alpha J(\vec{x}) \quad (\text{for all } \vec{x} \in \Omega, \text{ real number } \alpha, \alpha \vec{x} \in \Omega)$$

and the principle of additivity

$$J(\vec{x}_1 + \vec{x}_2) = J(\vec{x}_1) + J(\vec{x}_2) \quad (\text{for } \vec{x}_1, \vec{x}_2, \text{ and } \vec{x}_1 + \vec{x}_2 \in \Omega)$$

ex.) i) $J(x) = \int_{t_0}^{t_f} x(t) dt$; linear functional

Homogeneity

$$J(\alpha x) = \int_{t_0}^{t_f} \alpha x(t) dt = \alpha \int_{t_0}^{t_f} x(t) dt \Rightarrow J(\alpha x) = \alpha J(x)$$

Additivity

$$J(x_1) = \int_{t_0}^{t_f} x_1(t) dt, \quad J(x_2) = \int_{t_0}^{t_f} x_2(t) dt$$

$$J(x_1 + x_2) = \int_{t_0}^{t_f} [x_1(t) + x_2(t)] dt = \int_{t_0}^{t_f} x_1(t) dt + \int_{t_0}^{t_f} x_2(t) dt = J(x_1) + J(x_2)$$

ii) $J(x) = \int_{t_0}^{t_f} x^2(t) dt$; non-linear functional

Homogeneity?

$$\begin{aligned} J(\alpha x) &= \int_{t_0}^{t_f} \alpha^2 x^2(t) dt \\ &= \alpha^2 \int_{t_0}^{t_f} x^2(t) dt \\ &= \alpha^2 J(x) \end{aligned}$$

$$\Rightarrow J(\alpha x) \neq \alpha J(x)$$

<Increment of functional>

- Function f

$\vec{q}, \vec{q} + \Delta\vec{q}$ are elements of f

Then, increment of $f (\equiv \Delta f)$

$$\Delta f \equiv f(\vec{q} + \Delta\vec{q}) - f(\vec{q})$$

Δf depends on both $\vec{q}$ and $\Delta\vec{q} \rightarrow \Delta f(\vec{q}, \Delta\vec{q})$

- Functional J

$\vec{x}, \vec{x} + \delta\vec{x}$ are functions for J

$\delta\vec{x}$; the variation of the function $\vec{x}$

Then, increment of $J (\equiv \Delta J)$

$$\Delta J \equiv J(\vec{x} + \delta\vec{x}) - J(\vec{x})$$

ΔJ depends on both $\vec{x}$ and $\delta\vec{x} \rightarrow \Delta J(\vec{x}, \delta\vec{x})$

ex.)

$$\begin{aligned} J(x) &= \int_{t_0}^{t_f} x^2(t) dt \\ \Delta J &= J(x + \delta x) - J(x) \\ &= \int_{t_0}^{t_f} (x + \delta x)^2 dt - \int_{t_0}^{t_f} x^2 dt \\ &= \int_{t_0}^{t_f} [2x\delta x + (\delta x)^2] dt \end{aligned}$$

<The variation of functional>

- The increment of a function f of n variables

$$\Delta f(\vec{q}, \Delta \vec{q}) = \underbrace{df(\vec{q}, \Delta \vec{q})}_{\text{a linear function of } \Delta \vec{q}} + g(\vec{q}, \Delta \vec{q}) \cdot \|\Delta \vec{q}\|$$

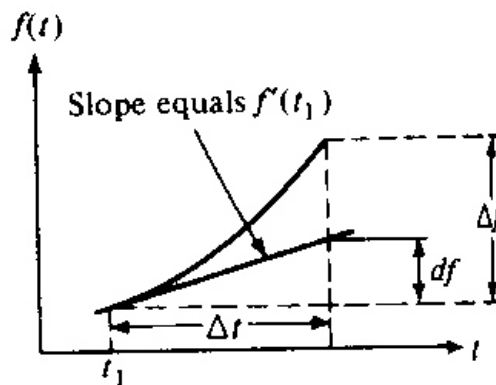
If $\lim_{\|\Delta \vec{q}\| \rightarrow 0} \{g(\vec{q}, \Delta \vec{q})\} = 0$, then, f is said to be **differentiable** at $\vec{q}$ and df is a

differential of f at the point $\vec{q}$.

If f is a differentiable function of one variable t , differential can be written

$$df(t, \Delta t) = f'(t) \Delta t \quad ; \quad \text{first-order approximation to } \Delta f$$

where $f'(t)$; derivative of f at t



- The increment of a functional

$$\Delta J(\vec{x}, \delta \vec{x}) = \underbrace{\delta J(\vec{x}, \delta \vec{x})}_{\text{linear term in } \delta \vec{x}} + g(\vec{x}, \delta \vec{x}) \cdot \|\delta \vec{x}\|$$

If $\lim_{\|\delta \vec{x}\| \rightarrow 0} \{g(\vec{x}, \delta \vec{x})\} = 0$

then, J is said to be **differentiable** on $\vec{x}$ and δJ is the **variation** of J evaluated for the function $\vec{x}$, δJ is the linear approximation to the difference in the functional J caused by $\vec{x}$, $\vec{x} + \delta \vec{x}$.

<Maxima and Minima of functional>

- A function f with domain $\mathbf{D}$ has a relative extremum at the point $\vec{q}^*$

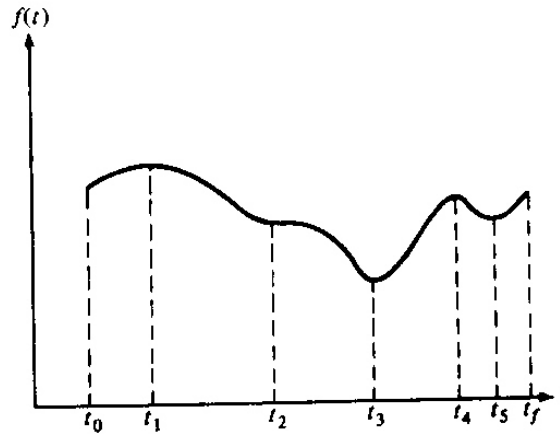
If there is $\varepsilon > 0$ ($\|\vec{q} - \vec{q}^*\| < \varepsilon$ for all point $\vec{q}$ in $\mathbf{D}$) such that the increment of f has the same sign.

$$f(\vec{q}^*) \text{ is a relative minimum} \quad \text{if } \Delta f = f(\vec{q}) - f(\vec{q}^*) > 0 \quad (\text{a})$$

$$f(\vec{q}^*) \text{ is a relative maximum} \quad \text{if } \Delta f = f(\vec{q}) - f(\vec{q}^*) < 0 \quad (\text{b})$$

If equation (a) is satisfied for arbitrarily large ε , then $f(\vec{q}^*)$ is a global or absolute minimum.

If equation (b) is satisfied for arbitrarily large ε , then $f(\vec{q}^*)$ is a global or absolute maximum.



stationary points	; t_1, t_2, t_3, t_4, t_5
extreme points	; t_1, t_3, t_4, t_5
relative minima	; t_0, t_3, t_5
relative maxima	; t_1, t_4, t_f
global minima	; t_3
global maxima	; t_1

• A functional J with domain Ω has a relative extremum at $\vec{x}^*$ (extremal), if there is an $\varepsilon > 0$ ($\|\vec{x} - \vec{x}^*\| < \varepsilon$ for all point $\vec{x}$ in Ω) such that the increment of J has the same sign.

$$J(\vec{x}^*) \text{ is a relative minimum if } \Delta J = J(\vec{x}) - J(\vec{x}^*) > 0 \quad (c)$$

$$J(\vec{x}^*) \text{ is a relative maximum if } \Delta J = J(\vec{x}) - J(\vec{x}^*) < 0 \quad (d)$$

If equation (c) is satisfied for arbitrarily large ε , then $J(\vec{x}^*)$ is a global or absolute minimum.

If equation (d) is satisfied for arbitrarily large ε , then $J(\vec{x}^*)$ is a global or absolute maximum.

Let $\vec{x}^*$ be the extremal and $J(\vec{x}^*)$ is the extremum, then $\Delta J(\vec{x}^*)$ must have the same sign for every $\vec{x}$ near $\vec{x}^*$ (i.e., every $\delta\vec{x}$).

If $\delta\vec{x}$ is sufficiently small,

$$\text{sign } \Delta J = \text{sign } \delta J$$

where ΔJ ; total variation

δJ ; first variation

$$\Delta J(\vec{x}, \delta\vec{x}) = \delta J(\vec{x}, \delta\vec{x}) + g(\vec{x}, \delta\vec{x}) \cdot \|\delta\vec{x}\|$$

linear term in $\delta\vec{x}$

$$\text{If } \lim_{\|\delta\vec{x}\| \rightarrow 0} \{g(\vec{x}, \delta\vec{x})\} \rightarrow 0, \quad \Delta J \approx \delta J$$

$$\begin{aligned} \Delta J(\vec{x}^*, \delta\vec{x}) &= J(\vec{x}^* + \delta\vec{x}) - J(\vec{x}^*) \\ &= J(\vec{x}^*) + \frac{\partial J}{\partial \vec{x}} \Big|_{\vec{x}^*} \delta\vec{x} + \frac{1}{2} \frac{\partial^2 J}{\partial \vec{x}^2} \Big|_{\vec{x}^*} (\delta\vec{x})^2 + \dots - J(\vec{x}^*) \\ &= \frac{\partial J}{\partial \vec{x}} \Big|_{\vec{x}^*} \delta\vec{x} + \frac{1}{2} \frac{\partial^2 J}{\partial \vec{x}^2} \Big|_{\vec{x}^*} (\delta\vec{x})^2 + H.O.T \end{aligned}$$

$$\text{where } \frac{\partial J}{\partial \vec{x}} \Big|_{\vec{x}^*} \delta\vec{x} \equiv \delta J(\vec{x}^*, \delta\vec{x}); \text{ first variation}$$

$$\frac{1}{2} \frac{\partial^2 J}{\partial \vec{x}^2} \Big|_{\vec{x}^*} (\delta\vec{x})^2 \equiv \delta^2 J; \text{ second variation}$$

δJ is a linear function of $\delta\vec{x}$

<The fundamental theorem of the calculus of variation>

- The fundamental theorem (necessary condition) used in finding extreme values of functions.

⇒ the differential must be zero at an extreme point (except extrema at the boundaries of closed region).

- In variational problems, the analogous theorem exists.

⇒ the variation must be zero on an extremal curve, if there are no bounds imposed on the curves.

Let $\vec{x}$ be a vector function of t in the class Ω .

$J(\vec{x})$ be a differentiable function of $\vec{x}$.

Assume $\vec{x}$ in Ω are not constrained by any boundaries.

∴ Fundamental theorem of the calculus of variations

If $\vec{x}^*$ is extremal and $J(\vec{x})$ is a smooth functional (differentiable) and $\vec{x}$ is unconstrained in the interior, the variation of J must vanish on $\vec{x}^*$ that is

$\delta J(\vec{x}^*, \delta \vec{x}) = 0$ for all admissible $\delta \vec{x}$.

<proof of contradiction>

Assume that $\vec{x}^*$ is an extremal and that $\delta J(\vec{x}^*, \delta \vec{x}) \neq 0$

→ the increment ΔJ can be made to change sign in an arbitrarily small neighborhood of $\vec{x}^*$.

→ contradiction of relative extremum for J at $\vec{x}^*$

; the increment of $J (= \Delta J)$ has the same sign at extremal $\vec{x}^*$.

(remember : Let $\vec{x}^*$ be the extremal and $J(\vec{x}^*)$ is the extremum, then $\Delta J(\vec{x}^*)$ must have the same sign for every $\vec{x}$ near $\vec{x}^*$.)

If $\delta \vec{x}$ is sufficiently small, $\text{sign } \Delta J = \text{sign } \delta J$

Proof in detail)

The increment is

$$\begin{aligned}\Delta J(\bar{x}^*, \delta \bar{x}) &= J(\bar{x}^* + \delta \bar{x}) - J(\bar{x}^*) \\ &= \delta J(\bar{x}^*, \delta \bar{x}) + g(\bar{x}^*, \delta \bar{x}) \cdot \|\delta \bar{x}\| \\ &\quad \text{as } \|\delta \bar{x}\| \rightarrow 0, \quad g(\bar{x}^*, \delta \bar{x}) \rightarrow 0\end{aligned}$$

$$\Rightarrow \Delta J \approx \delta J$$

① Let us select the variation

$$\begin{aligned}\delta \bar{x}(t) &= \alpha \delta \bar{x}^{(1)}(t) \\ \alpha &> 0 \text{ and } \|\alpha \delta \bar{x}^{(1)}\| < \varepsilon\end{aligned}$$

suppose that

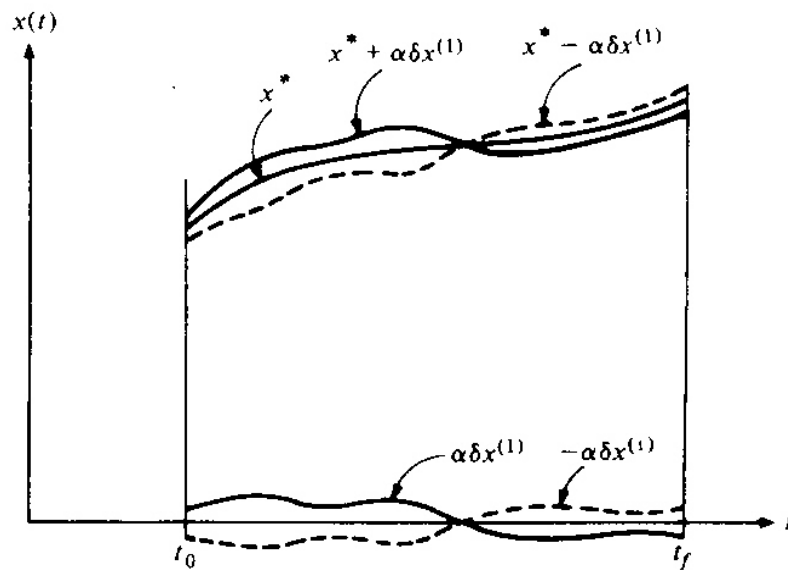
$$\delta J(\bar{x}^*, \alpha \delta \bar{x}^{(1)}) < 0 \quad (\text{Note } \delta J(\bar{x}^*, \delta \bar{x}) = \left. \frac{\partial J}{\partial \bar{x}} \right|_{\bar{x}^*} \delta \bar{x})$$

Since δJ is a linear functional of $\delta \bar{x}$, the principle of homogeneity gives

$$\delta J(\bar{x}^*, \alpha \delta \bar{x}^{(1)}) = \left. \frac{\partial J}{\partial \bar{x}} \right|_{\bar{x}^*} \alpha \delta \bar{x}^{(1)} = \alpha \delta J(\bar{x}^*, \delta \bar{x}^{(1)}) < 0 \Rightarrow \delta J(\bar{x}^*, \delta \bar{x}^{(1)}) < 0$$

$$\Rightarrow \delta J(\bar{x}^*, \alpha \delta \bar{x}^{(1)}) = \delta J(\bar{x}^*, \delta \bar{x}) < 0 \quad (\text{note } \delta \bar{x} = \alpha \delta \bar{x}^{(1)})$$

$$\rightarrow \Delta J(\bar{x}^*, \delta \bar{x}) < 0$$



② Let us select the variation

$$\delta \vec{x}(t) = -\alpha \delta \vec{x}^{(l)}(t)$$

$$\alpha > 0, \quad \left\| -\alpha \delta \vec{x}^{(l)} \right\| < \varepsilon$$



The principle of homogeneity

$$\delta J(\vec{x}^*, -\alpha \delta \vec{x}^{(l)}) = -\alpha \delta J(\vec{x}^*, \delta \vec{x}^{(l)}) > 0 \quad (\text{since } \delta J(\vec{x}^*, \delta \vec{x}^{(l)}) < 0)$$

$$\Rightarrow \delta J(\vec{x}^*, -\alpha \delta \vec{x}^{(l)}) = \delta J(\vec{x}^*, \delta \vec{x}) > 0 \quad (\text{note } \delta \vec{x} = -\alpha \delta \vec{x}^{(l)})$$

$$\Rightarrow \Delta J(\vec{x}^*, \delta \vec{x}) > 0$$



③ from ① and ②

If $\delta J(\vec{x}^*, \delta \vec{x}) \neq 0$

then, $\Delta J(\vec{x}^*, \alpha \delta \vec{x}^{(l)}) = \Delta J(\vec{x}^*, \delta \vec{x}) < 0$

where $\delta \vec{x} = \alpha \delta \vec{x}^{(l)}$, because $\delta \vec{x}$ can be arbitrary near $\vec{x}^*$

$$\Delta J(\vec{x}^*, -\alpha \delta \vec{x}^{(l)}) = \Delta J(\vec{x}^*, \delta \vec{x}) > 0$$



where $\delta \vec{x} = -\alpha \delta \vec{x}^{(l)}$, because $\delta \vec{x}$ can be arbitrary near $\vec{x}^*$

→ contradiction of the assumption that $\vec{x}^*$ is extremal (then, ΔJ has the same sign)

→ Hence, if $\vec{x}$ is an extremal, it is necessary that $\delta J(\vec{x}^*, \delta \vec{x}) = 0$ for arbitrary $\delta \vec{x}$ when $\vec{x}$ in Ω are not bounded.